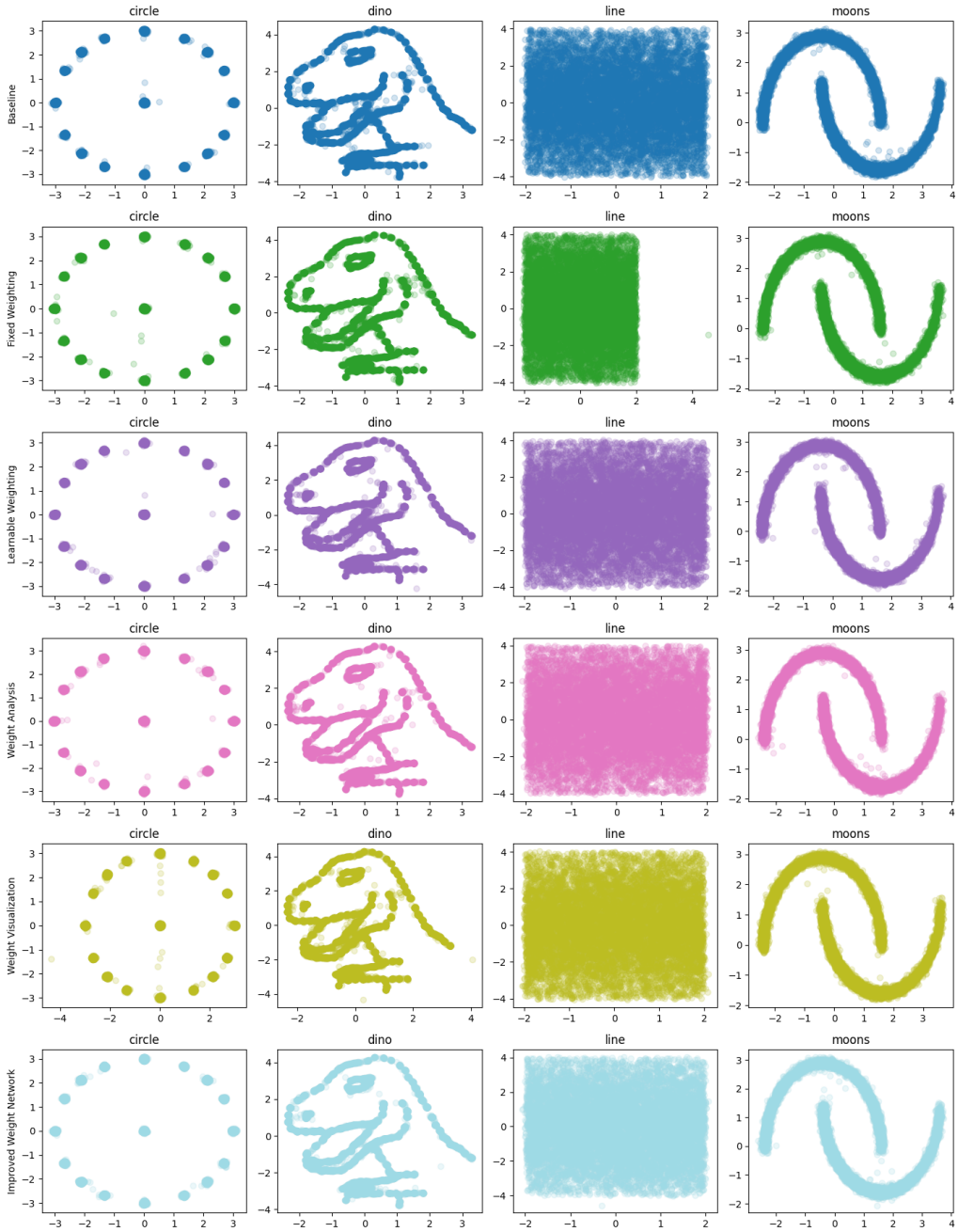


Figure 1: Generated samples from our adaptive dual-scale diffusion model across different runs and datasets. Each row represents a different experimental run, while columns show results for circle, dino, line, and moons datasets.

2.2 Adaptive Mechanisms in Generative Models

Adaptive mechanisms have been explored in various contexts within generative modeling. The Time-dependent Multihead Self Attention (TMSA) mechanism introduced in DiffiT Hatamizadeh et al. (2023) demonstrates the potential of adaptive, time-dependent processing in diffusion models. While conceptually similar in its time-dependent nature, our approach differs in its specific focus on balancing multi-scale features in low-dimensional spaces, rather than attention mechanisms in

high-dimensional data. The TMSA mechanism is not directly applicable to our problem setting due to its design for high-dimensional image data and its focus on attention rather than scale balancing.

Bai et al. (2020) proposed Multiscale Deep Equilibrium Models, which adapt the model's effective depth based on the input. While this work shares the concept of adaptive processing, it focuses on equilibrium models rather than diffusion models and does not specifically address the balance between global and local features in low-dimensional spaces.

Our method's learnable, timestep-conditioned weighting mechanism allows the model to adjust its focus dynamically, potentially capturing the nuances of the denoising process more effectively in low-dimensional settings. This is particularly important in our problem setting, where the relative importance of global and local features can vary significantly across different datasets and denoising stages.

2.3 Low-dimensional Diffusion Models

While much of the research on diffusion models has focused on high-dimensional data such as images, there is growing interest in applying these models to low-dimensional spaces. TabDDPM Kotelnikov et al. (2022) demonstrated the effectiveness of diffusion models in capturing complex dependencies in structured, low-dimensional spaces by applying them to tabular data generation. However, TabDDPM does not specifically address the challenge of balancing global structure and local details, which is the primary focus of our work.

Our approach extends this line of research by introducing an adaptive dual-scale method specifically designed to improve the fidelity and quality of generated samples in low-dimensional spaces. Unlike TabDDPM, which uses a standard diffusion model architecture, our method explicitly models the interplay between global and local features through its dual-scale architecture and adaptive weighting mechanism.

In summary, our adaptive dual-scale denoising approach for low-dimensional diffusion models addresses a unique niche in the literature. While it builds upon foundations laid by previous work in multi-scale and adaptive processing, it is specifically tailored to the challenges of low-dimensional spaces. Our method's dynamic balancing of global and local features sets it apart from fixed multi-scale approaches and makes it particularly suited for capturing complex low-dimensional distributions. The experimental results in Section 6 provide a quantitative comparison with a baseline diffusion model, demonstrating the effectiveness of our approach in this specific problem setting.

3 Background

Diffusion models have emerged as a powerful class of generative models, achieving remarkable success in various domains of machine learning Yang et al. (2023). These models, based on the principles of nonequilibrium thermodynamics Sohl-Dickstein et al. (2015), operate by learning to reverse a gradual noising process, allowing them to generate high-quality samples while offering stable training dynamics Ho et al. (2020).

The diffusion process consists of two main phases:

1. Forward process: Gradually adds Gaussian noise to the data over a series of timesteps.
2. Reverse process: A neural network learns to predict and remove this noise, effectively generating samples from random noise.

Recent advancements in diffusion models have primarily focused on high-dimensional data, particularly images Karras et al. (2022b). However, the study of diffusion models in low-dimensional spaces remains crucial for:

- Providing tractable analysis of model behavior, informing improvements in higher-dimensional settings.
- Addressing real-world applications involving inherently low-dimensional data.
- Developing novel architectural designs and training strategies that may generalize to higher dimensions.

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3.1 Problem Setting

We focus on applying diffusion models to 2D datasets. Let $\mathcal{X} \subset \mathbb{R}^2$ be our data space, and $p_{\text{data}}(\mathbf{x})$ be the true data distribution over $\mathcal{X}$. Our goal is to learn a generative model that samples from a distribution $p_{\text{model}}(\mathbf{x})$ closely approximating $p_{\text{data}}(\mathbf{x})$.

The diffusion process is defined over T timesteps. Let $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ be a sample from the data distribution, and $\mathbf{x}_1, \ldots, \mathbf{x}_T$ be the sequence of increasingly noisy versions of $\mathbf{x}_0$. The forward process is defined as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \tag{1}$$

where β_t is the noise schedule.

The reverse process, parameterized by a neural network ϵ_θ, is defined as:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)) \tag{2}$$

In low-dimensional spaces, each dimension carries significant information about the overall structure of the data. This presents a unique challenge: the model must simultaneously capture both the global structure and local details of the data distribution. Traditional diffusion models often struggle to achieve this balance in low dimensions, motivating our proposed adaptive dual-scale approach.

Our approach is based on two key assumptions:

1. The importance of global and local features varies across different datasets and at different stages of the denoising process.
2. A learnable, timestep-conditioned weighting mechanism can effectively balance the contributions of global and local features during denoising.

These assumptions form the basis of our adaptive dual-scale denoising architecture, which we will describe in detail in the following section.

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4 Method

Our adaptive dual-scale denoising approach addresses the challenge of balancing global structure and local details in low-dimensional diffusion models. Building upon the formalism introduced in Section 3, we present a novel architecture that dynamically adjusts its focus between global and local features throughout the denoising process.

4.1 Dual-Scale Architecture

The core of our method is a dual-scale architecture that processes the input at two different scales simultaneously:

1. Global Scale: This branch processes the original input $\mathbf{x}_t \in \mathcal{X} \subset \mathbb{R}^2$, capturing the overall structure of the data.
2. Local Scale: This branch processes an upscaled version of the input $\mathbf{x}_t^{up} \in \mathbb{R}^4$, focusing on fine-grained details.

Both branches use similar network architectures, but with different input dimensions:

$$\epsilon_\theta^{\text{global}}(\mathbf{x}_t, t) = \text{MLP}_{\text{global}}(\mathbf{x}_t, t) \tag{3}$$

$$\epsilon_\theta^{\text{local}}(\mathbf{x}_t^{up}, t) = \text{MLP}_{\text{local}}(\mathbf{x}_t^{up}, t) \tag{4}$$

where MLP denotes a multi-layer perceptron with sinusoidal embeddings for both input and time, similar to the architecture used in the original DDPM Ho et al. (2020). The upscaling operation $\mathbf{x}_t^{up} = \text{Upscale}(\mathbf{x}_t)$ is implemented as a learnable linear transformation: